Lesson 8: Capstone: Battle Arena

Module 2 | 60 minutes | game design, functions, data | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Your mission
Design and build a turn-based text battle game where your hero fights a random enemy using attack and heal, with stats in dictionaries and a function for every action.

Checklist

- [ ] Player and enemy stats in dictionaries, with a list of at least 3 enemies
- [ ] Functions for damage and healing that return numbers within limits
- [ ] Random damage and a randomly chosen enemy
- [ ] A game loop that runs while both fighters have HP
- [ ] attack and heal moves, limited potions and an Unknown move message
- [ ] Victory and defeat endings, then GAME OVER

Milestones

1. Design sheet done: hero, 3 enemies with HP and damage, and moves — done on: _____
2. apply_damage and heal_hp tested with fixed numbers — done on: _____
3. One full turn works: stats, move, enemy hits back — done on: _____
4. The full battle loop ends with a win or a loss; a friend has playtested and you balanced one number — done on: _____

Plan it

Write the steps of your program in plain English (pseudocode) before you code.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Program works	Crashes or never ends	One turn works but the battle does not end properly	Full battles end with the right winner every time	Works perfectly and handles odd input and empty potions
Data and functions	No dictionaries or functions	Some stats in variables; one function	Stats in dictionaries, enemy list and helper functions that return values	Clean functions (or a class) with no repeated code and clear names
Game design and balance	No choices for the player	Attack only; the game is always won or always lost	Attack and heal with random damage; the game feels fair	Extra moves or enemies, and balance changes explained from playtests
Testing and presentation	Never tested; no comments	Tested once; few comments	Helpers tested and code commented; clear messages each turn	Playtested by others, improved, and presented with a short design explanation

Want more?

- Add a Player class with attack and heal methods
- Add a special move that does big damage but can miss
- Fight three enemies in a row and count the wins

Two stars and a wish

Ask a friend to test your project.

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